

NEWBOLD, PA, USA

WMO: 997698

Lat: 40.137N

Lon: 74.752W

Elev: 18

StdP: 101.11

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.3	(c) -8.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 11.1	(k) -4.0	(l) 10.1	(m) -3.7	(n) 4.2	(o) 310	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.8	(c) 33.0	(d) N/A	(e) 31.5	(f) N/A	(g) 30.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.2	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.6	(b) 7.5	(c) 6.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	0.5	1.7	6.0	12.6	18.3	22.6	25.6	24.2	20.6	14.7	8.3	4.4		
	DBStd	9.50	5.33	4.77	4.61	4.28	3.87	3.14	2.52	2.36	3.18	4.08	4.00	4.61		
	HDD10.0	961	297	236	140	18	0	0	0	0	0	8	79	183		
	HDD18.3	2508	553	466	383	181	49	4	0	1	13	126	302	431		
	CDD10.0	2185	2	4	16	96	257	378	482	441	319	153	27	11		
	CDD18.3	692	0	0	0	9	48	132	224	183	83	13	0	0		
	CDH23.3	5234	0	0	1	81	341	976	2004	1311	465	55	0	0		
	CDH26.7	1681	0	0	0	18	77	303	767	394	113	8	0	0		
(14) Wind	WSAvg	2.6	3.2	2.9	3.1	2.9	2.6	2.5	2.2	2.1	2.1	2.5	2.7	2.8		
(15) Precipitation	PrecAvg	1074	79	75	91	79	87	87	125	100	106	72	79	94		
	PrecMax	1509	228	131	186	168	158	159	340	245	230	138	197	177		
	PrecMin	772	9	25	24	21	6	16	33	12	20	1	8	15		
	PrecStd	193	47	31	39	35	43	45	81	62	58	36	50	47		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	18.7	21.9	29.4	31.2	33.6	35.6	33.5	32.1	27.5	20.9	19.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	14.6	14.8	18.7	25.7	28.9	31.6	33.7	31.6	29.7	24.8	18.8	16.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	11.2	10.9	15.6	22.9	26.9	30.0	32.0	30.3	27.7	22.5	16.5	14.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	8.4	8.2	13.2	20.3	25.2	28.2	30.7	29.0	25.9	20.7	14.7	11.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	7.7	8.5	9.9	9.4	8.9	8.8	8.2	8.7	8.1	8.1	6.8		
		MCDBR	11.5	11.9	12.4	13.5	11.6	10.3	10.2	9.6	10.7	10.2	11.2	9.6		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.321	0.359	0.389	0.436	0.500	0.497	0.490	0.404	0.361	0.329	0.309		
	taud		2.466	2.431	2.385	2.329	2.226	2.088	2.111	2.118	2.332	2.463	2.510	2.524		
	Ebn at Noon		854	896	893	885	846	790	788	781	834	840	826	830		
	Edh at Noon		78	93	109	123	139	160	155	150	114	90	75	69		
(44) All-Sky Solar Radiation	RadAvg		1.92	2.83	3.75	4.84	5.36	5.88	6.01	5.19	4.32	3.07	2.18	1.66		
	RadStd		0.17	0.30	0.30	0.43	0.61	0.50	0.29	0.34	0.46	0.31	0.19	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+68

Nomenclature: See separate page